

## HERSTMONCEUX OBS, United Kingdom

WMO: 038840

Lat: 50.8700N

Lon: 0.3466E

Elev: 17

StdP: 101.12

Time Zone: 0.00 (EUW)

Period: 82-92

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -4.4       | -2.7 | -8.3                          | 1.9 | -3.2 | -6.3 | 2.2 | -0.6 | 17.8                  | 10.1 | 15.3 | 8.6  | 3.4                   | 60   | 0.617 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.4                    | 24.8            | 18.4 | 23.2 | 17.4 | 21.7 | 16.7 | 19.1                | 23.8 | 18.2 | 21.9 | 17.4 | 20.5 | 4.1                  | 70   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 17.5                            | 12.5 | 20.7 | 16.7 | 11.9 | 19.6 | 16.1 | 11.5 | 19.0 | 54.3          | 23.9 | 51.4 | 21.9 | 49.0 | 20.6 | 24.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 13.7          | 12.0 | 10.4 | DB                         | -7.0 | 28.4               | 4.0 | 2.4                                                | -9.8  | 30.1       | -12.1 | 31.5       | -14.3 | 32.9       | -17.2 | 34.6 | (4) |
| (5)               |      |      | WB                         | -7.5 | 21.6               | 4.0 | 1.5                                                | -10.3 | 22.7       | -12.7 | 23.6       | -14.9 | 24.5       | -17.8 | 25.6 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 10.1        | 4.6   | 4.0   | 6.2   | 8.0   | 11.5  | 14.0  | 16.7  | 16.4  | 14.2  | 11.6  | 7.6   | 6.2   | (6)  |  |
|                                                                        | DBStd       | 5.23        | 3.54  | 3.63  | 2.58  | 2.65  | 2.98  | 2.66  | 2.24  | 2.37  | 2.36  | 2.63  | 3.42  | 3.13  | (7)  |  |
|                                                                        | HDD10.0     | 774         | 169   | 169   | 121   | 72    | 16    | 1     | 0     | 0     | 1     | 14    | 88    | 123   | (8)  |  |
|                                                                        | HDD18.3     | 3028        | 427   | 402   | 377   | 311   | 213   | 134   | 63    | 67    | 125   | 208   | 323   | 378   | (9)  |  |
|                                                                        | CDD10.0     | 812         | 0     | 1     | 2     | 11    | 62    | 120   | 207   | 199   | 127   | 65    | 16    | 4     | (10) |  |
|                                                                        | CDD18.3     | 24          | 0     | 0     | 0     | 0     | 1     | 3     | 11    | 9     | 1     | 0     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 142         | 0     | 0     | 0     | 0     | 4     | 18    | 62    | 54    | 3     | 0     | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 22          | 0     | 0     | 0     | 0     | 0     | 1     | 6     | 14    | 0     | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 4.3         | 5.3   | 4.9   | 4.7   | 4.4   | 3.9   | 3.8   | 3.6   | 3.7   | 3.6   | 4.3   | 4.2   | 4.9   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 804         | 91    | 62    | 46    | 52    | 49    | 51    | 51    | 61    | 61    | 91    | 101   | 88    | (15) |  |
|                                                                        | PrecMax     | 1158        | 185   | 150   | 133   | 114   | 106   | 156   | 116   | 134   | 127   | 267   | 234   | 160   | (16) |  |
|                                                                        | PrecMin     | 601         | 19    | 7     | 5     | 5     | 10    | 11    | 9     | 3     | 6     | 16    | 36    | 10    | (17) |  |
|                                                                        | PrecStd     | 132         | 42    | 38    | 24    | 31    | 27    | 37    | 26    | 35    | 36    | 52    | 53    | 45    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 11.6  | 12.8  | 14.4  | 19.5  | 23.8  | 25.5  | 27.7  | 28.5  | 23.6  | 19.6  | 15.2  | 13.3  | (19) |  |
|                                                                        |             | MCWB        | 10.2  | 11.1  | 10.8  | 12.6  | 16.5  | 18.6  | 20.3  | 19.2  | 17.6  | 15.8  | 13.4  | 11.5  | (20) |  |
|                                                                        | 2%          | DB          | 11.0  | 11.1  | 12.3  | 16.3  | 21.5  | 23.0  | 25.0  | 24.3  | 21.2  | 17.7  | 14.2  | 12.4  | (21) |  |
|                                                                        |             | MCWB        | 9.9   | 9.5   | 9.6   | 11.6  | 15.8  | 17.5  | 18.7  | 17.8  | 16.7  | 15.1  | 12.9  | 10.9  | (22) |  |
|                                                                        | 5%          | DB          | 10.3  | 10.0  | 11.0  | 14.4  | 19.1  | 21.1  | 23.3  | 22.5  | 19.7  | 16.6  | 13.5  | 11.6  | (23) |  |
|                                                                        |             | MCWB        | 9.3   | 8.7   | 8.8   | 10.3  | 14.5  | 16.4  | 17.7  | 17.0  | 15.8  | 14.3  | 12.3  | 10.4  | (24) |  |
|                                                                        | 10%         | DB          | 9.6   | 9.0   | 10.0  | 12.5  | 16.8  | 19.0  | 21.6  | 21.0  | 18.5  | 15.7  | 12.7  | 10.9  | (25) |  |
|                                                                        |             | MCWB        | 8.6   | 7.7   | 8.4   | 9.5   | 13.0  | 15.1  | 16.7  | 16.6  | 15.3  | 13.8  | 11.6  | 9.9   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 10.9  | 11.1  | 11.5  | 13.9  | 17.8  | 19.8  | 20.8  | 20.0  | 18.6  | 16.7  | 13.8  | 12.1  | (27) |  |
|                                                                        |             | MCDB        | 11.2  | 12.2  | 13.8  | 17.4  | 22.8  | 23.9  | 26.2  | 25.5  | 22.0  | 18.8  | 14.6  | 12.8  | (28) |  |
|                                                                        | 2%          | WB          | 10.2  | 9.9   | 10.2  | 12.1  | 16.2  | 18.1  | 19.4  | 18.7  | 17.6  | 15.5  | 13.1  | 11.2  | (29) |  |
|                                                                        |             | MCDB        | 10.6  | 10.7  | 11.5  | 15.5  | 20.3  | 22.1  | 23.9  | 23.5  | 20.0  | 17.0  | 14.0  | 12.0  | (30) |  |
|                                                                        | 5%          | WB          | 9.5   | 8.9   | 9.5   | 11.0  | 14.7  | 16.8  | 18.3  | 17.8  | 16.9  | 14.9  | 12.5  | 10.6  | (31) |  |
|                                                                        |             | MCDB        | 10.1  | 9.7   | 10.6  | 13.5  | 18.3  | 20.5  | 22.1  | 21.0  | 18.7  | 16.1  | 13.4  | 11.4  | (32) |  |
|                                                                        | 10%         | WB          | 8.8   | 8.0   | 8.8   | 10.0  | 13.4  | 15.4  | 17.3  | 17.2  | 16.1  | 14.2  | 11.8  | 9.9   | (33) |  |
|                                                                        |             | MCDB        | 9.6   | 8.9   | 9.8   | 12.1  | 16.7  | 18.3  | 20.4  | 20.0  | 18.0  | 15.4  | 12.7  | 10.8  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 5.1   | 5.1   | 6.4   | 7.5   | 8.0   | 7.8   | 8.4   | 8.4   | 8.1   | 6.5   | 5.7   | 5.0   | (35) |  |
|                                                                        |             | MCDBR       | 5.2   | 5.4   | 7.2   | 10.0  | 11.2  | 11.1  | 11.4  | 10.6  | 9.6   | 6.9   | 4.4   | 4.6   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 5.0   | 4.7   | 5.2   | 6.2   | 6.8   | 6.7   | 6.0   | 5.3   | 5.7   | 4.8   | 3.9   | 4.2   | (37) |  |
|                                                                        |             | MCWBR       | 5.3   | 5.0   | 6.1   | 8.5   | 10.3  | 10.1  | 9.9   | 8.9   | 8.1   | 5.9   | 4.5   | 4.5   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.335 | 0.352 | 0.390 | 0.408 | 0.415 | 0.414 | 0.409 | 0.402 | 0.384 | 0.367 | 0.340 | 0.325 | (40) |  |
|                                                                        |             | taud        | 2.364 | 2.363 | 2.256 | 2.232 | 2.269 | 2.298 | 2.319 | 2.358 | 2.394 | 2.412 | 2.406 | 2.356 | (41) |  |
|                                                                        | Ebn at Noon |             | 679   | 771   | 800   | 830   | 842   | 844   | 829   | 803   | 743   | 669   | 626   | 57    | (42) |  |
|                                                                        |             | Edh at Noon | 65    | 84    | 111   | 126   | 127   | 125   | 120   | 110   | 96    | 79    | 62    | 57    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 0.82  | 1.53  | 2.72  | 4.37  | 5.20  | 5.77  | 5.51  | 4.60  | 3.42  | 1.94  | 0.96  | 0.63  | (44) |  |
|                                                                        | RadStd      |             | 0.07  | 0.21  | 0.33  | 0.38  | 0.35  | 0.48  | 0.45  | 0.32  | 0.23  | 0.18  | 0.11  | 0.08  | (45) |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (71 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page